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Adaptive Spatial-Temporal Window: Unlocking the Potential of Event Cameras in Heterogeneous Velocity Scenarios

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Abstract

Most event-based algorithms typically split the event stream into fixed groups (e.g., fixed time or fixed count) for downstream processing, lacking adaptivity to scene dynamics. Several adaptive partitioning strategies have been proposed, but they are unable to cope well with heterogeneous velocity scenarios (HVS) involving both fast- and slow-moving objects. To address this issue, we propose Adaptive Spatial-Temporal Window (ASTW) strategy, which simultaneously achieves temporal adaptivity and spatial locality in event partitioning. Based on the principle of maximum entropy, we derive a patch-level time window determination criterion and efficiently implement it based on event density and vectorized calculations. Experiments on publicly available event-based object detection and tracking datasets demonstrate that ASTW significantly outperforms existing state-of-the-art partitioning strategies. We also construct HetVel, the first RGB-event dual-modality dataset for HVS, and further highlight the advantages of ASTW on this challenging benchmark. We believe that our ASTW strategy and the constructed HetVel dataset will advance the field of neuromorphic vision.

1. Introduction

Event cameras [10, 35] are bio-inspired vision sensors with advantages such as high temporal resolution, low latency, and high dynamic range [19], which have demonstrated strong potential across a wide range of vision tasks [4, 12, 15, 31, 34]. In event-based vision tasks, event streams are typically converted into suitable event representations for subsequent processing. This process typically consists of two steps: (i) partitioning the event stream into different groups, and (ii) transforming them into representations using the representation strategy. Considerable research has been conducted on representation strategies [2, 5, 6, 11, 17, 23, 30, 38]. In contrast, research on partitioning strategies remains relatively limited, and is the main focus of our work. The most commonly used partitioning methods are Fixed Time Window [23] and Fixed Number of Events [28]. However, such fixed strategies rely on predetermined partitioning rules that cannot adapt to scene dynamics, and apply a uniform processing scheme across the entire field of view (FOV), thereby lacking spatial locality.

Several adaptive partitioning strategies have been proposed, but they typically focus only on temporal adaptivity while neglecting spatial locality [7, 8, 18, 25, 26, 37]. For scenarios containing only a single object whose velocity varies over time (i.e., temporal heterogeneity), existing adaptive strategies can handle the situation reasonably well. However, in more common scenarios where multiple objects in the FOV exhibit distinct motion velocities (i.e., spatial heterogeneity), existing adaptive strategies struggle to perform effectively due to the lack of spatial locality. These scenarios involving temporal heterogeneity and spatial heterogeneity are collectively referred to as heterogeneous velocity scenarios (HVS) [13, 16]. Among the partitioning strategies that account for spatial locality, however, they either lack adaptivity [3] or introduce high computational overhead [24]. Therefore, it is particularly important to develop a computationally efficient event partitioning strategy that simultaneously ensures both spatial locality and temporal adaptivity.

In this work, we thus propose an Adaptive Spatial-temporal Window (ASTW) strategy, which enables a computationally efficient event partitioning method incorporating both spatial locality and temporal adaptivity. To introduce spatial locality, we divide the pixel plane into non-overlapping patches and regard each patch as the fundamental spatial unit. Based on the principle of maximum entropy, the partitioning strategy should simultaneously satisfy spatial consistency constraint and local information maximization: the former ensures a balanced distribution of information throughout the FOV, while the latter guarantees that each local patch contains sufficient information. Building

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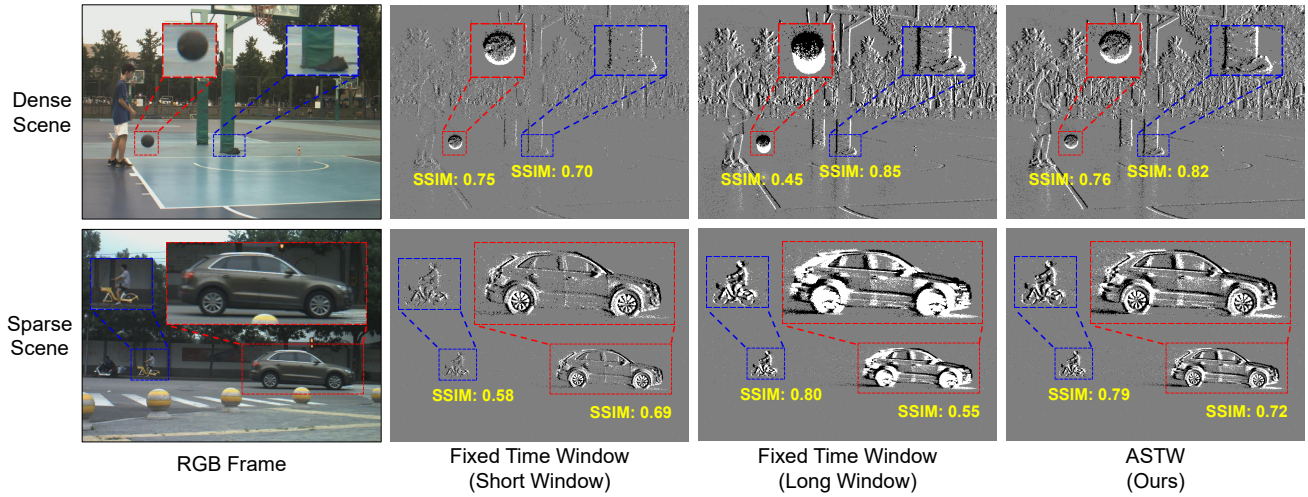


Figure 1. Visualization of Fixed Time Window (baseline) and our proposed Adaptive Spatial-Temporal Window (ASTW) in HVS. Red and blue boxes highlight fast- and slow-moving objects, respectively, with zoom-in views. ASTW strategy maintains a high structural similarity index measure (SSIM) with RGB images among objects with varying velocities, verifying its temporal adaptivity and spatial locality.

on the above constraints, we further derive an analytical formula for determining the patch-level time window based on event density to reduce the computational overhead on velocity estimation. To enhance the robustness and multi-scale adaptivity of event density, we apply spatial smoothing to the event density and fuse historical information over time using Exponential Moving Average (EMA). To maintain causal consistency of the scene, we propose a minimum time-step strategy that unifies the temporal reference across all patches and achieves progressive updates of the partitioning process. A vectorized implementation is employed to avoid per-patch iterative computations, reducing the algorithm's computational complexity to $\mathcal{O}(N)$.

We validate our ASTW strategy on both object detection and object tracking tasks. We conduct experiments on different network architectures (recurrent and feedforward) and backbone networks (CNNs and Transformers), and the results consistently demonstrate that applying the ASTW strategy leads to performance improvements. Compared to the second-best results, ASTW achieves improvements of up to 2.6 mAP and 2.0 AP_{50} on the Gen1 dataset, as well as gains of 0.2 SR, comparable PR, and 0.3 NPR on the EventVOT dataset. Given the lack of event-based benchmark datasets for temporal and spatial heterogeneity, we construct HetVel, the first RGB-event dual-modality dataset focusing on HVS. On this challenging dataset, the advantage of the ASTW strategy over other partitioning methods is further highlighted. Through case studies, we intuitively demonstrate the temporal adaptivity and spatial locality of the ASTW strategy. Experiments show that our ASTW strategy greatly reduces the latency introduced by the partitioning process while effectively handling motion speeds of up to 2400 px/s. Our contributions are summarized as follows:

- We propose Adaptive Spatial-temporal Window (ASTW) strategy that simultaneously ensures both spatial locality and temporal adaptivity in event partitioning.
- We derive an analytical formula for determining the patch-level time window based on the principle of maximum entropy, and efficiently implement it using event density and vectorized calculations.
- Experiments on object detection and tracking tasks show that our ASTW strategy significantly outperforms state-of-the-art (SOTA) partitioning strategies.
- We construct HetVel, the first RGB-Event dual-modality dataset designed for HVS, further highlighting the advantages of ASTW.

2. Related Work

In event-based vision, HVS can be handled at the representation level. Methods such as SITS [22] and VK-SITS [1] adapt event representations according to local velocity to produce more consistent feature appearances. However, they still rely on upstream partitioning to determine the time window, making it difficult to simultaneously achieve low-latency responses for fast objects and sufficient information accumulation for slow objects. As an orthogonal approach, event partitioning offers greater flexibility and potential for handling HVS. Therefore, this section primarily focuses on the research progress of event partitioning strategies.

A common approach is to partition event data into non-overlapping slices using Fixed Time Window [23], thereby

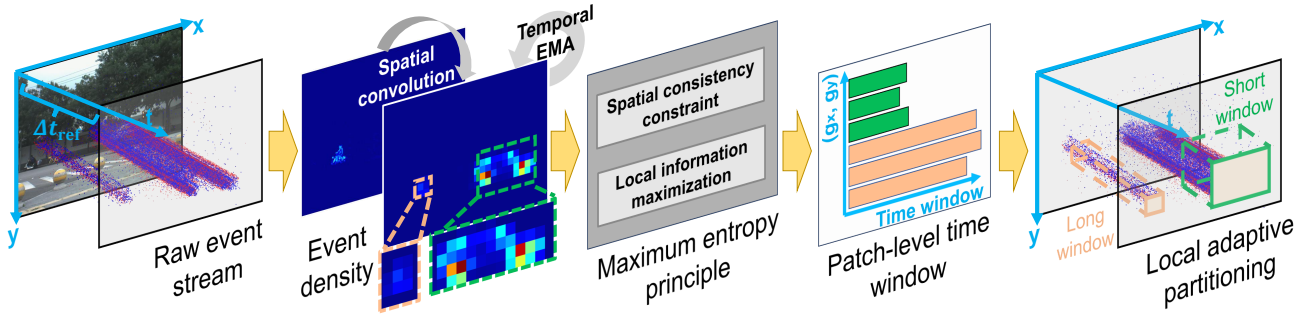


Figure 2. Workflow of Adaptive Spatial-Temporal Window strategy. First, count the number of events within a reference time window and calculate the event density for each patch. Then, calculate the patch-level time window length based on the event density. Finally, achieve local and adaptive partitioning through an efficient vectorized implementation.

generating a fixed-frame-rate video. Another approach is to partition event data using Fixed Number of Events [28], which ensures the consistent size of data blocks and improves computational efficiency in sparse scenes. However, both fixed strategies cannot adapt to scene dynamics and typically require parameter tuning. These methods are straightforward and normally used as baselines.

To achieve temporal adaptivity, several adaptive strategies have been proposed. Adaptive Time-Surface with Linear Time Decay (ATSLTD) strategy [8] measures the information content of the current frame and triggers partitioning when it reaches a predefined confidence interval. Adaptive Temporal Sampling (ATS) strategy [18] searches for the optimal time window such that the number of events within the window approximates a dynamic target value. Adaptive Global Decay strategy [25] integrates scene dynamics into the decay process, enabling an adaptive partitioning of the event stream. Adaptive Event Conversion (AEC) strategy [26] partitions the event stream when the number of events within a given time window exceeds a predefined threshold, while streams that do not reach the threshold continue to accumulate. SpikeSlicer [7] employs a spiking neural network (SNN) as a dynamic trigger that emits spikes at moments corresponding to optimal downstream task performance, thereby achieving event stream partitioning. Although the aforementioned strategies are adaptive, they lack spatial locality, which is also crucial for event partitioning.

Although some strategies consider spatial locality, they each still have certain limitations. Time-Ordered Recent Event (TORE) strategy [3] constructs queues of length k at each pixel location, which ensures locality but lacks adaptivity. Event Lifetime [24] estimates per-event velocity to infer its lifetime by incorporating both locality and adaptivity, but it suffers from high computational overhead and limited robustness. Moreover, TORE and Event Lifetime only select events at a given moment, without providing a mechanism for dynamically updating the partitioning process.

Table 1. Comparison of different partitioning strategies.

Partitioning Strategy	Temporal Adaptivity	Spatial Locality
Fixed Time Window [23]	✗	✗
Fixed Number of Events [28]	✗	✗
ATSLTD [8]	✓	✗
Adaptive Temporal Sampling [18]	✓	✗
Adaptive Event Conversion [26]	✓	✗
Adaptive Global Decay [25]	✓	✗
SpikeSlicer [7]	✓	✗
TORE [3]	✗	✓
Event Lifetime [24]	✓	✓
ASTW (Ours)	✓	✓

We summarize the common partitioning strategies and their characteristics in Tab. 1. In summary, existing partitioning strategies either fail to achieve both adaptivity and locality simultaneously, or introduce high computational overhead. Our proposed ASTW strategy incorporates both adaptivity and locality. Based on spatial consistency constraint and local information maximization, we derive an analytical formula for determining the patch-level time window. This formula reveals that the selection of the time window depends on edge length and motion velocity. By introducing a statistics-based event density, we avoid explicitly estimating these quantities and achieved efficient computation through vectorized implementation.

3. Method

In this section, we first model the event data triggered by motion and identify the core task in handling HVS as determining an appropriate time window for each spatial patch. We then introduce two constraints based on the maximum entropy principle and further derive an analytical formula for determining the patch-level time window. Finally, we present an efficient vectorized implementation that achieves acceptable computational complexity and latency. The overall workflow of the ASTW strategy is illustrated in Fig. 2.

3.1. Motion-Triggered Event Modeling

Unlike frame-based cameras, the event camera outputs a sparse, asynchronous event stream. An event is triggered when the logarithmic change in brightness at a pixel location exceeds a predefined threshold. Each event e_k can be described by (x_k, y_k, t_k, p_k), where (x_k, y_k) denotes the pixel coordinates, t_k denotes the timestamp, and p_k represents the polarity, indicating whether the brightness increases or decreases. In our modeling, we adopt the brightness constancy assumption [32], which implies that events are triggered solely by object motion.

For cases where there is only a single object in the FOV, the relationship between the amount of event data and motion can be expressed as

$$N(\Delta t) = c \cdot L \cdot v \cdot \Delta t, \tag{1}$$

where $N(\Delta t)$ denotes the number of events generated within the time window Δt, c denotes the coefficient of object edge contrast, L and v denote the effective edge length of the object and its average velocity, respectively.

For cases involving heterogeneous velocities, employing a fixed strategy (e.g., Fixed Time Window) is insufficient to comprehensively represent all objects. Fast- and slow-moving objects tend to choose time windows of different lengths for the clearest appearance, as shown in Fig. 1. To enable differentiated processing of various spatial regions, we divide the pixel plane into non-overlapping patches to introduce spatial locality. For ease of analysis, we consider the edge contrast c as a constant. The core task is to find the most appropriate time window length Δt for each patch to better present the appearance of objects in HVS.

3.2. Adaptive Spatial-Temporal Window

Based on the principle of maximum entropy, we impose two constraints for the partitioning process:

Spatial Consistency Constraint: Ensures that the amount of information is uniform across different patches, reduce the non-uniformity caused by spatial differences.

Local Information Maximization: Encourages each patch to carry as much information as possible, providing rich data for subsequent feature extraction and pattern recognition.

The partitioned event data typically need to be further transformed into event representations to serve as algorithm input. Here we adopt the simplest Event Frame [27] as event representation, which is a binary image, with 1 and 0 indicating the presence or absence of an event at each pixel location, respectively. For a single patch, if all pixels are 0 or 1, it carries low information content; ideally, the patch should depict a clear edge of the object, with a mix of 1 and 0. The information entropy [29] of a patch can be expressed as

$$H_{ij} = -p_{ij} \cdot \log_2 p_{ij} - (1 - p_{ij}) \cdot \log_2 (1 - p_{ij}), \tag{2}$$

where H_{ij} denotes the information entropy of the patch indexed by (i, j), and p_{ij} represents the proportion of nonzero pixels within that patch. Considering a patch size of (s, s), p_{ij} equals the area swept by the object edge divided by the patch area, i.e., $p_{ij} = L_{ij} \cdot v_{ij} \cdot \Delta t_{ij} / s^2$. When $p_{ij} = 1/2$, the derivative $H'_{ij} = 0$, and H_{ij} attains its maximum value of 1 bit.

According to **Local Information Maximization**, achieving the maximum at a specific patch requires $p = 1/2$. In conjunction with **Spatial Consistency Constraint**, we therefore aim for all patches to satisfy $p_{ij} = 1/2$. We then obtain

$$\frac{c \cdot L_{ij} \cdot v_{ij} \cdot \Delta t_{ij}}{s^2} = c \cdot \frac{1}{2} = \gamma, \tag{3}$$

where γ is an information constant. From Eq. 3, it can be seen that the longer the object's edge length within a patch and the faster its motion, the shorter the required time window should be.

For convenience, we define event density D_{ij} as

$$D_{ij} = \frac{N_{ij}}{\Delta t_{\text{ref}} \cdot s^2} = \frac{c \cdot L_{ij} \cdot v_{ij}}{s^2}, \tag{4}$$

where N_{ij} denotes the number of events in spatial patch (s, s) and reference time window Δt_{ref}. The second equality is derived from Eq. 1, indicating that the event density reflects the combined influence of motion velocity and edge length. Substituting Eq. 4 into Eq. 3 yields $D_{ij} \cdot \Delta t_{ij} = \gamma$. Thus, by continuously calculating D_{ij} using the first half of Eq. 4, then the time window $\Delta t_{ij} = \gamma / D_{ij}$.

The event density defined in Eq .4 lacks robustness in describing object motion characteristics, particularly when the object exhibits diverse spatial scales (potentially spanning multiple patches) or undergoes rapid motion variations over time. Therefore, we adopt spatial smoothing and temporal smoothing to correct the event density. Specifically, we use a convolution operator with a size of $n \cdot s$ to smooth the event density map, thereby combining information from different spatial scales. We adopt the Exponential Moving Average (EMA) method to fuse historical information over time. The corrected event density is expressed as

$$\begin{cases} \widetilde{D}_{ij} = (K * D)_{ij} \\ \bar{D}_{ij}(t) = \alpha \cdot \widetilde{D}_{ij}(t) + (1 - \alpha) \cdot \bar{D}_{ij}(t - 1), \end{cases} \tag{5}$$

where K denotes the convolution kernel, and α represents the smoothing factor. Fig. 3 presents the representations generated based on D_{ij} and $\bar{D}_{ij}$, respectively. It can be observed that applying spatial and temporal smoothing effectively suppresses artifacts at the boundaries.

Accordingly, Eq.3 can be rewritten as

$$\Delta t_{ij} = \text{clamp}(\frac{\gamma}{\bar{D}_{ij} + \varepsilon}, \Delta t_{\text{min}}, \Delta t_{\text{max}}), \tag{6}$$

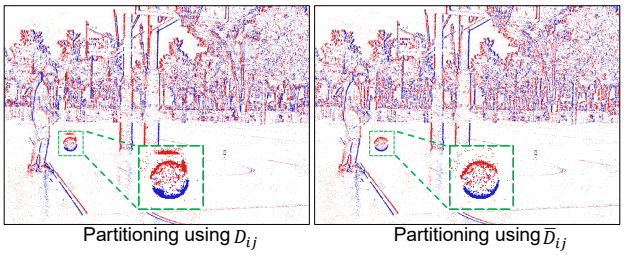


Figure 3. Visualization comparison before and after spatial and temporal smoothing.

where clamp function constrains Δt_{ij} between Δt_{min} and Δt_{max}, and ε is a small constant (set to 1×10^{-3}) used to prevent division by zero.

To prevent patch-level time window partitioning from violating the causal consistency of the scene, we propose a minimum time-step strategy that unifies the temporal reference across all patches and achieves progressive updates of the partitioning process. Specifically, all patches share the same ending timestamp, and each patch extracts events forward according to its corresponding time window Δt_{ij}. The global timestamp is updated based on the smallest window among all patches. This strategy ensures that no events are missed during the partitioning process. This can result in overlapping time windows for some patches. This was found acceptable through experiments. The workflow of ASTW strategy is summarized in Algorithm 1 and Fig. 2.

3.3. Vectorized Implementation

In Algorithm 1, the efficiency-critical step is Line 6, where P_{cand} is extracted from $\mathcal{E}$ using Δt_{ij}: $P_{\text{cand}} \leftarrow \{e_k \in \mathcal{E} \mid T_{global} - \Delta t_{ij} \leq t_k < T_{global}\}$. To avoid patch-by-patch loops, we employ an efficient vectorized implementation. First, all event coordinates (x, y) are mapped to patch indices (g_x, g_y) by integer division with the patch size s. Then, using (g_x, g_y) as indices, a vectorized lookup in Δt_{ij} retrieves the corresponding time window for each event. Finally, compare the timestamp of each event with the retrieved time window and retain events that fall in the time window.

This fully vectorized implementation achieves a time complexity of $\mathcal{O}(N)$. Statistical results show that our ASTW strategy exhibits good computational efficiency, with an average latency of 7.43 μs/event, which is on the same order of magnitude as TORE (2.04 μs/event). It outperforms Event Lifetime (56.38 μs/event) and Adaptive Global Decay (16.76 μs/event) by factors of 7.6 $\times$ and 2.3 $\times$, respectively. Detailed complexity analysis and statistical results are provided in Supplementary Material S1.

Algorithm 1 ASTW Strategy

Input: Event stream $\mathcal{E}$ ($e_k \in \mathcal{E}, k = 1, 2, \ldots, N$), global timestamp T_{global}, list of labels $\mathcal{L}$ ($L_m \in \mathcal{L}, m = 1, 2, \ldots, M$)
Output: List of partitioned events $\mathcal{P}$ ($P_m \in \mathcal{P}, m = 1, 2, \ldots, M$)

1: Set reference time window Δt_{ref} and patch size (s, s):
2: **while** $T_{\text{global}} < e_N(t)$ **or not** EOF **do**
3: Calculate D_{ij} within $[T_{\text{global}} - \Delta t_{\text{ref}}, T_{\text{global}})$, Eq. 4
4: Calculate $\widetilde{D}_{ij}$ and $\bar{D}_{ij}$, Eq. 5
5: Calculate time window Δt_{ij} for each patch, Eq. 6
6: Extract candidate partition P_{cand} from $\mathcal{E}$ using Δt_{ij}
7: Choose minimum time window and update T_{global}:

$$\Delta t_{\text{step}} = \min(\Delta t_{ij}), T_{\text{global}} \leftarrow T_{\text{global}} + \Delta t_{\text{step}}$$

8: **if** $T_{\text{global}} \leq L_{m+1}(t)$ **and** $T_{\text{global}} > L_m(t)$ **then**
9: $P_m \leftarrow P_{\text{cand}}, m \leftarrow m + 1$
10: Save partitioned events P_m and label L_m
11: **end if**
12: **end while**

4. Experiments

We evaluate the ASTW strategy on two publicly available datasets and our proposed HetVel dataset. The experiments are carried out on a computational platform equipped with 8 NVIDIA RTX 3090 GPUs and an Intel Xeon Gold 6130 CPU.

We compare our ASTW strategy with commonly used partitioning strategies discussed in Related Work. For fair comparison, we either adopt the parameter settings recommended in the original papers, or perform hyperparameter search to obtain their optimal values. The implementation details of each strategy can be found in Supplementary Material S2. During the experiment, we ensure that the representation strategy, training configurations, and network model are completely consistent, with only the partitioning strategy being different.

4.1. Event-Based Object Detection

The Gen1 Automotive Detection dataset [9] is an event-based object detection dataset, containing over 39 hours of recorded data captured by an Asynchronous Time-based Image Sensor (ATIS) at a resolution of 304×240. The dataset provides bounding box annotations for two object classes, cars and pedestrians, with annotation frequencies of 1, 2, or 4 Hz.

We conduct experiments on both feedforward and recurrent models. For the feedforward models, we adopt the two-stage object detection framework, Faster R-CNN. We employ ResNet-50 [14] and Swin Transformer V2 [21] as the backbones, combined with a Feature Pyramid Network (FPN) [20]. During training, we use the AdamW optimizer with an initial learning rate of 5×10^{-4}, a weight decay

Table 2. Comparison of task performance metrics of different strategies. The left half of the table presents the experimental results for the object detection task, while the right half displays the results for the object tracking task. The best performance is highlighted in **bold** and dark green.

Partitioning Strategy	Object Detection						Object Tracking					
	ResNet-50		Swin V2		RVTs		EventVOT Dataset			HetVel Dataset		
	mAP	AP_{50}	mAP	AP_{50}	mAP	AP_{50}	SR	PR	NPR	SR	PR	NPR
Fixed Time Window [23]	47.9	74.5	43.9	73.2	46.9	73.4	61.7	60.2	84.3	47.7	56.1	81.6
Fixed Number of Events [28]	47.5	74.5	44.0	73.1	46.2	72.4	60.1	56.7	82.4	48.1	55.9	82.2
Adaptive Temporal Sampling [18]	47.4	74.3	44.0	72.6	45.9	72.1	60.8	57.5	83.5	48.2	55.7	82.4
Adaptive Event Conversion [26]	45.0	71.1	41.1	69.7	43.6	69.2	58.0	52.2	80.3	44.9	51.9	78.6
Adaptive Global Decay [25]	46.0	73.9	41.6	69.9	47.1	73.6	60.8	57.7	83.5	48.4	56.1	82.2
SpikeSlicer [7]	46.9	73.4	42.9	72.8	45.6	71.5	61.9	60.5	84.4	48.4	56.0	82.4
TORE [3]	48.3	75.2	43.6	73.1	47.3	73.8	62.1	58.7	84.8	48.6	56.3	82.7
Event Lifetime [24]	45.2	72.0	40.8	72.1	44.7	69.8	58.7	53.4	81.1	46.0	53.4	79.8
ASTW (Ours)	**50.6**	**76.0**	**46.6**	**75.2**	**48.3**	**75.4**	**62.3**	**60.5**	**85.1**	**50.7**	**58.5**	**84.9**

coefficient of 1×10^{-4}, a batch size of 12, and a total of 4 training epochs. For the recurrent models, we adopt Recurrent Vision Transformers (RVTs) [12], which employ LSTM units to accumulate temporal information. RVTs integrate self-attention modules with LSTM units, and the detection head is based on the YOLOX framework. The detailed experimental settings and hyperparameters are kept consistent with [12].

For visualization, we convert the partitioned events into event representations, overlapped with the inference results and corresponding labels (Fig. 4). It demonstrates that the ASTW strategy, through its local and adaptive partitioning, effectively accommodates varying motion speeds, enabling clearer object contours in the scene. Furthermore, the prediction errors in terms of false positives and false negatives are reduced with the ASTW strategy, particularly for small objects.

We summarize the comparison results of the partitioning strategies using mean Average Precision (mAP) and AP_{50} as evaluation metrics. As shown in the left half of Tab. 2, our strategy achieves the best performance across different model configurations, i.e., 50.6 / 46.6 mAP and 76.0 / 75.2 AP_{50} with ResNet-50 and Swin V2 backbones in feedforward models, and 48.3 mAP and 75.4 AP_{50} in recurrent models. Compared with the second-best results, ASTW improves by 2.6 mAP and 2.0 AP_{50}. Notably, feedforward models exhibit larger performance gains than recurrent models in our ASTW strategy. We attribute this to the fact that recurrent architectures inherently learn temporal dependencies during training, which reduces their reliance on the partitioning process.

The Method section suggests that objects with different motion speeds should choose different time windows, and here we further validate this claim through experiments. We associate independent ground-truth bounding boxes into continuous trajectories using an IoU-based Hungarian matching algorithm, and then estimate each object's speed. The samples are then divided into low, medium, and high speed groups. Fig. 5 shows that the proposed ASTW strategy achieves consistent performance improvements across different motion speeds, further highlighting the importance of spatial locality in the partitioning process.

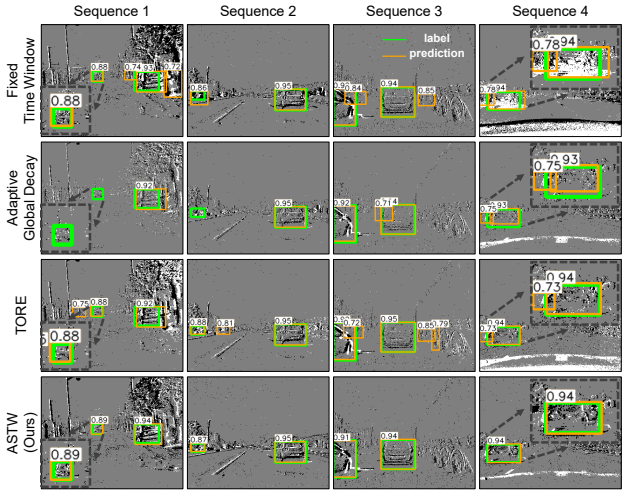


Figure 4. Visualization of event representations on the Gen1 dataset, overlapped with the inference results and corresponding labels. Some details are shown using zoom-in views. Predicted bounding boxes with a confidence score below 0.7 have been removed.

4.2. Event-Based Object Tracking

EventVOT dataset [34] is an event-based object tracking dataset, shot by a Prophesee camera EVK-HD at a resolution of 1280×720, containing a total of 1141 videos. For model selection, we employ a specialized model designed for event data, HDETrack [34], to evaluate the effective-

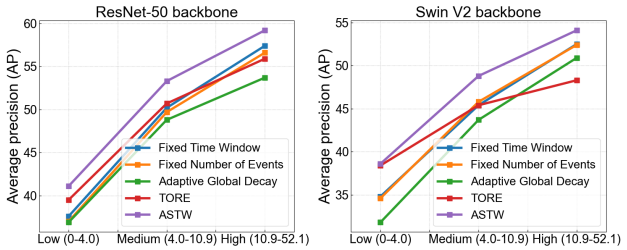


Figure 5. Detection performance under different motion speeds.

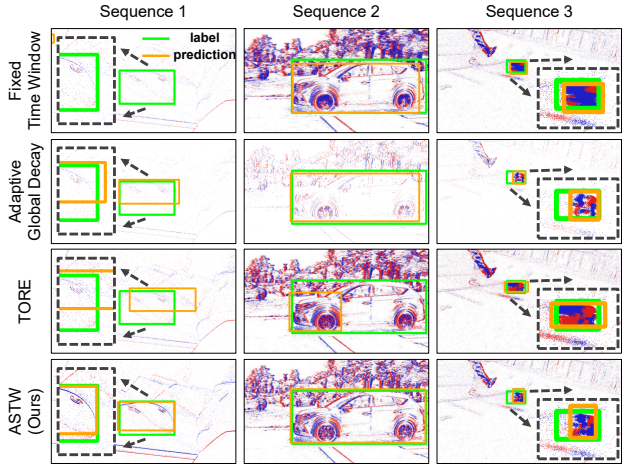


Figure 6. Visualization of event representations on the EventVOT dataset, overlapped with the inference results and corresponding labels. Some details are shown using zoom-in views.

ness of different partitioning strategies in object tracking tasks. HDETrack is a hierarchical knowledge distillation framework that employs a teacher–student paradigm, transferring knowledge from a multimodal (RGB–event) teacher network to an event-only student tracker. The model architecture is built upon a unified Transformer backbone, similar to OSTrack [36] and CEUTrack [33].

We visualize the event representations and inference results corresponding to different partitioning strategies (Fig. 6). The three sequences in Fig. 6 illustrate typical scenarios involving slow motion, fast motion, and their simultaneous presence. The visualization results indicate that our ASTW strategy is motion-adaptive and consistently provides sharp object contours, enabling more robust and accurate object tracking.

We summarize the comparison results of the partitioning strategies using Success Rate (SR), Precision Rate (PR), and Normalized Precision Rate (NPR) as evaluation metrics. As shown in the right half of Tab. 2, our method achieves the best results: SR 62.3, PR 60.5, and NPR 85.1. TORE and Fixed Time Window strategy also achieve comparable performance. We speculate that this is because

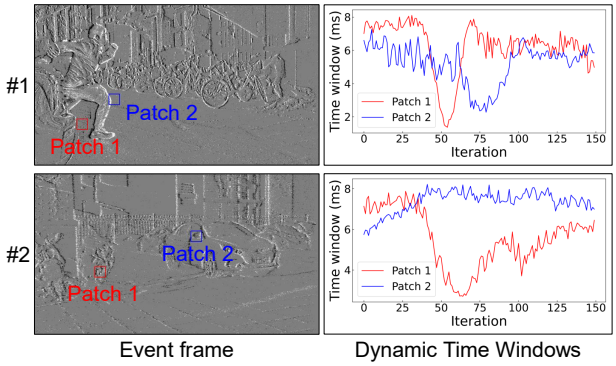


Figure 7. Visualization of event frames and dynamic time windows in two typical scenarios. The results indicate that our ASTW strategy can differentially process distinct spatial patches and adaptively adjust the time window length in response to scene dynamics.

tracking tasks rely more on feature response strength, and therefore have lower sensitivity to partitioning strategies compared to detection tasks. We observe that although Event Lifetime simultaneously provides locality and adaptivity, it does not exhibit clear performance advantages. This is probably related to velocity estimation, which is prone to errors.

4.3. HetVel Dataset Evaluation

We introduce HetVel, the first RGB-Event dual-modality dataset designed for HVS. We set up a dual-modality data acquisition device, equipped with a Teledyne FLIR Blackfly-S as the frame-based (RGB) camera and a Prophesee EVK-4 as the event camera. The RGB frames serve as an auxiliary modality for generating accurate labels at a high frame rate (100 frames per second, fps). The HetVel dataset consists of 33 RGB-event videos, covering HVS in common scenes such as traffic and sports. A detailed description and visualization of the HetVel dataset can be found in Supplementary Material S3.

We select the object tracking task for validation. The HDETrack model, trained on the EventVOT dataset, is fine-tuned on the HetVel dataset. We randomly divide the HetVel dataset into 20 videos for fine-tuning and 13 videos for testing. The experimental results are presented in the right half of Tab. 2. Our proposed ASTW strategy achieves significantly superior performance across all evaluation metrics, yielding improvements of 2.1 SR, 2.2 PR, and 2.2 NPR over the second-best method.

To demonstrate the locality and adaptivity of the ASTW strategy, we select two typical scenarios as case studies: sudden pedestrian appearances in traffic scenarios and rapid saccadic motion in robot scenarios.

In the field of autonomous driving, timely responses to

objects that suddenly enter the field of view (e.g., pedestrians) are crucial. However, global partitioning paradigms typically require accumulation of events, which hinders rapid reactions to sudden intrusions. As illustrated in Fig. 7, our proposed ASTW strategy exhibits spatial locality, able to immediately shorten the time window for rapid response and high-frequency updates. Experiments show that in scenes with fast-moving objects, ASTW can adaptively partition a 1-second event stream into approximately 800–1000 temporal slices, generating an output every 1 ms on average. Estimates indicate that a vehicle traveling at 100 km/h moves only 0.03 m during a single slice, greatly reducing the latency introduced by the partitioning process.

Another scenario is rapid saccadic motion, which is commonly encountered for robots scanning their surroundings and identifying regions of interest. As shown in Fig. 7, our ASTW strategy enables adaptive partitioning, preserving clear contour information in the scene even during fast motion. In contrast, non-adaptive partitioning methods are prone to motion blur and require careful parameter selection to meet dynamic scene conditions. Experiments demonstrate that our ASTW strategy can effectively handle motion speeds of 2400 px/s.

4.4. Ablation Study

Ablation studies are performed on the object detection Gen1 dataset, using ResNet-50 as the backbone. The ablation results are shown in Tab. 3: removing patch division and causal consistency leads to mAP drops of 4.0 and 1.2, respectively, indicating that spatial locality and causal consistency are crucial. Further removing spatial smoothing and temporal smoothing results in mAP decreases of 0.5 and 0.3, demonstrating that these modules also make practical contributions. Tab. 4 shows that, compared with Fixed Time Window (baseline), our method achieves consistent performance improvements across different event representations.

We further analyze the parameter sensitivity of ASTW strategy, and the results are shown in Tab. 5-8. A patch size of 4 achieves the best performance, providing a favorable balance between statistical stability and spatial locality. For the information constant γ, we test its value in 1.0-2.0, and the resultant mAP remains stable, so we use 1.7. We test the reference time window Δt_{ref} within 50-750 ms, the performance is stable within 250-500 ms with a slight decrease beyond this range, so we use 250 ms. Regarding Δt_{min} and Δt_{max}, we test multiple combinations. The results show that the combination $\Delta t_{\text{min}} = 10$ ms and $\Delta t_{\text{max}} = 250$ ms yields the best results. Further analysis can be found in Supplementary Material S4.

5. Conclusion

In this work, we propose Adaptive Spatial-Temporal Window (ASTW) strategy, which efficiently achieves event partitioning with temporal adaptivity and spatial locality. Based on the principle of maximum entropy, we derive an analytical formula for determining the patch-level time window and provide an efficient implementation using event density and vectorization calculations. Experiments on event-based object detection and tracking show that our strategy substantially outperforms existing partitioning methods. We construct HetVel, the first RGB-Event dual-modality dataset designed for heterogeneous velocity scenarios. Our ASTW strategy demonstrates more pronounced advantages on this challenging benchmark. We believe that our ASTW strategy and the constructed HetVel dataset will advance the field of neuromorphic vision.

Table 3. Component analysis of ASTW strategy.

Patch Division	Spatial Smoothing	Temporal Smoothing	Causal Consistency	mAP
✗	✓	✓	NA	46.6
✓	✗	✓	✓	50.1
✓	✓	✗	✓	50.3
✓	✓	✓	✗	49.4
✓	✓	✓	✓	**50.6**

Table 4. Experiments on different event representations.

Representation	Time Surface [30]	Event Count [23]	Voxel Grid [38]
Baseline	51.3	50.1	47.6
ASTW	**53.0**	**52.1**	**49.0**

Table 5. Sensitivity of ASTW strategy to patch size.

Patch Size	1	2	4	8	16	32	64
mAP	49.9	50.4	**50.6**	50.3	50.0	49.3	48.9

Table 6. Sensitivity of ASTW strategy to information constant.

Information Constant	1.0	1.3	1.5	1.7	2.0
mAP	50.4	50.5	50.3	**50.6**	50.0

Table 7. Sensitivity of ASTW strategy to reference time window.

Reference Time Window	50	100	250	500	750
mAP	49.4	50.1	**50.6**	**50.6**	50.0

Table 8. Sensitivity of the ASTW strategy to $(\Delta t_{\text{min}}, \Delta t_{\text{max}})$.

$(\Delta t_{\text{min}}, \Delta t_{\text{max}})$	(2.5, 250)	(5, 250)	(10, 250)	(25, 150)
mAP	50.0	50.1	**50.6**	49.5
$(\Delta t_{\text{min}}, \Delta t_{\text{max}})$	(25, 200)	(50, 150)	(50, 200)	(75, 150)
mAP	49.6	47.8	48.7	48.8

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